

QUANTUM SCATTERING REPORT

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1. Part 1 - Free Gaussian wavepacket

The numerical calculations use the lecture convention $m=1$ and $\hbar=1$. The initial Gaussian wavepacket is

$$\psi(0, x) = \left(\frac{2\alpha}{\pi}\right)^{\frac{1}{4}} \exp\left(-\alpha(x - x_0)^2\right) \exp\left(ip_0(x - x_0)\right)$$

For the free-particle case, $V(x)=0$. With $m=\hbar=1$, the stationary Schrödinger equation reduces to

$$-\frac{1}{2} \frac{d^2 u_p(x)}{dx^2} = E u_p(x).$$

Substituting $E=p^2/2$ gives $d^2 u_p(x)/dx^2 + p^2 u_p(x) = 0$, so the solutions are plane waves $\exp(\pm ipx)$. The normalized momentum eigenfunction used for the expansion is

$$u_p(x) = \frac{1}{\sqrt{2\pi}} \exp(ipx).$$

This plane-wave basis is then used to decompose the initial Gaussian wavepacket. The corresponding momentum-space coefficient $\phi(p)$ is obtained from

$$\phi(p) = \frac{1}{\sqrt{2\pi}} \int \psi(0, x) \exp(-ipx) dx$$

Under free evolution, the coefficient $\phi(p)$ remains unchanged while each momentum component acquires the phase $\exp(-ip^2 t/2)$. Hence the wavepacket at time t is

$$\psi(t, x) = \frac{1}{\sqrt{2\pi}} \int \phi(p) \exp(ipx) \exp(-ip^2 t/2) dp$$

1.1 Initial Gaussian: changing α

At fixed mean momentum $p_0=1$, increasing α decreases the width of the wavepacket in real space, since $\Delta x = 1/(2\sqrt{\alpha})$. Therefore, the Gaussian becomes narrower as α increases.

initial gaussians - width is changing

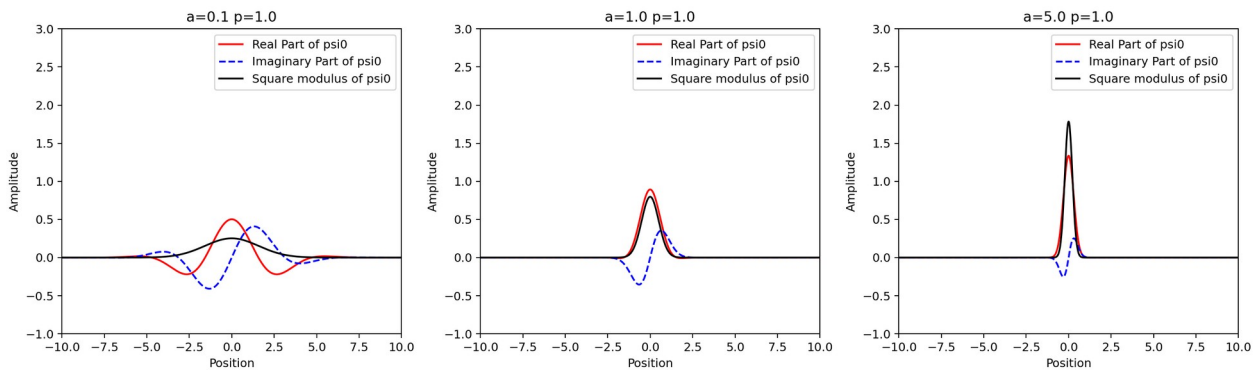


Figure 1. Initial Gaussian states for $\alpha=0.1, 1$, and 5 at fixed $p_0=1$. The black curve is $|\psi|^2$; red/blue curves are the real/imaginary parts.

1.2 Initial Gaussian: changing mean momentum p_0

At fixed $\alpha=1$, as p_0 increases, the wavefunction oscillates more rapidly, while the probability density $|\psi|^2$ remains unchanged.

initial gaussians - momentum is changing

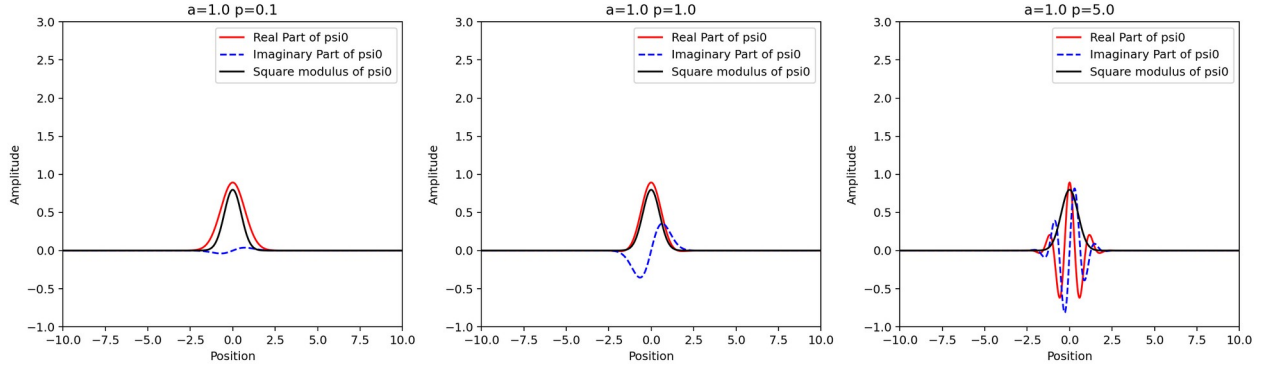


Figure 2. Initial Gaussian states for $p_0=0.1, 1$, and 5 at fixed $\alpha=1$. The phase oscillates more rapidly as p_0 increases.

1.3 Free propagation and spreading ($t>0$)

The propagation formula above is used for the time snapshots. As time increases, the center of the free Gaussian wavepacket moves to the right with velocity p_0/m . At the same time, the wavepacket spreads in real space, so its spatial width increases continuously during propagation.

free wavepacket propagation - snapshots

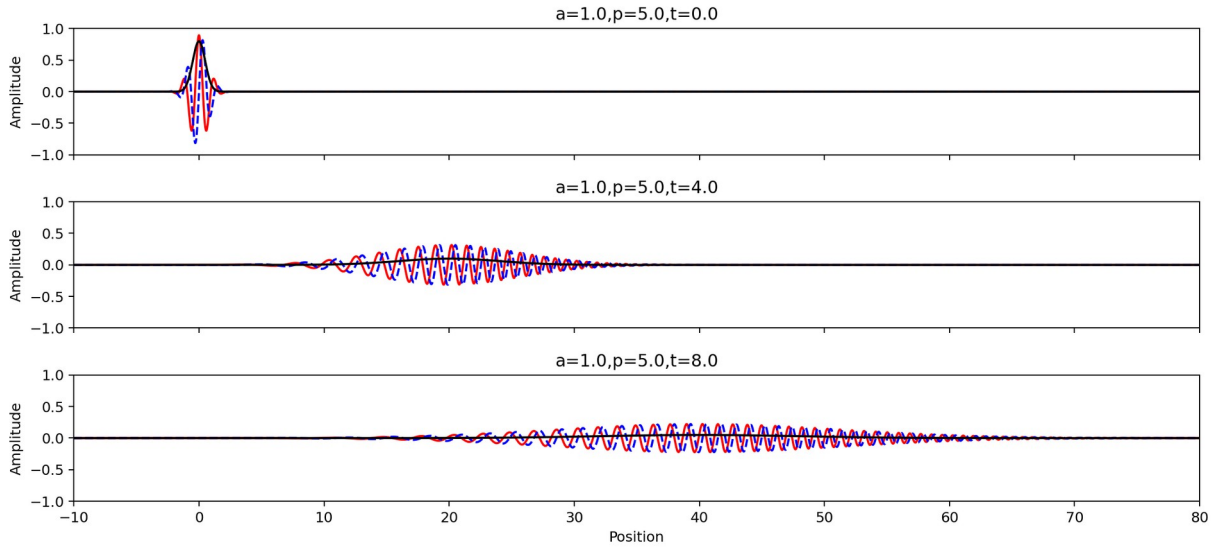


Figure 3. Free propagation snapshots for $\alpha=1$, $p_0=5$. The wavepacket moves to the right at the expected group velocity and broadens with time.

2. Part 2 - Regularized Dirac delta function

For a smooth test function $f(p)$, the Dirac delta is characterized by

$$\int_{-\infty}^{+\infty} f(p) \delta(p) dp = f(0).$$

The finite-L regularization used in the numerical calculation is

$$\delta_L(p) = \frac{1}{2\pi} \int_{-L}^L \exp(ipx) dx = \frac{\sin(pL)}{\pi p}.$$

In the large-L limit, this regularized form reproduces the defining action of the Dirac delta:

$$\lim_{L \rightarrow \infty} \int_{-\infty}^{+\infty} f(p) \delta_L(p) dp = f(0).$$

2.1 Numerical distribution test

To check this behavior numerically, $f(p)$ is taken as the momentum-space Gaussian $\phi(p)$ from Part 1, with $\alpha=1$ and $p_0=5$. The integration window is kept fixed at $p \in [-10, 10]$ for all L .

L	$\int f(p) \delta_L(p) dp$	$f(0)$	Absolute difference
10	0.0012166507	0.0012193111	2.660e-06
100	0.0012190874	0.0012193111	2.237e-07
1000	0.0012193480	0.0012193111	3.696e-08

The error decreases from 2.66×10^{-6} at $L=10$ to 3.70×10^{-8} at $L=1000$. The convergence toward $f(0)=0.0012193111$ is a direct numerical verification of the defining action of the Dirac delta distribution.

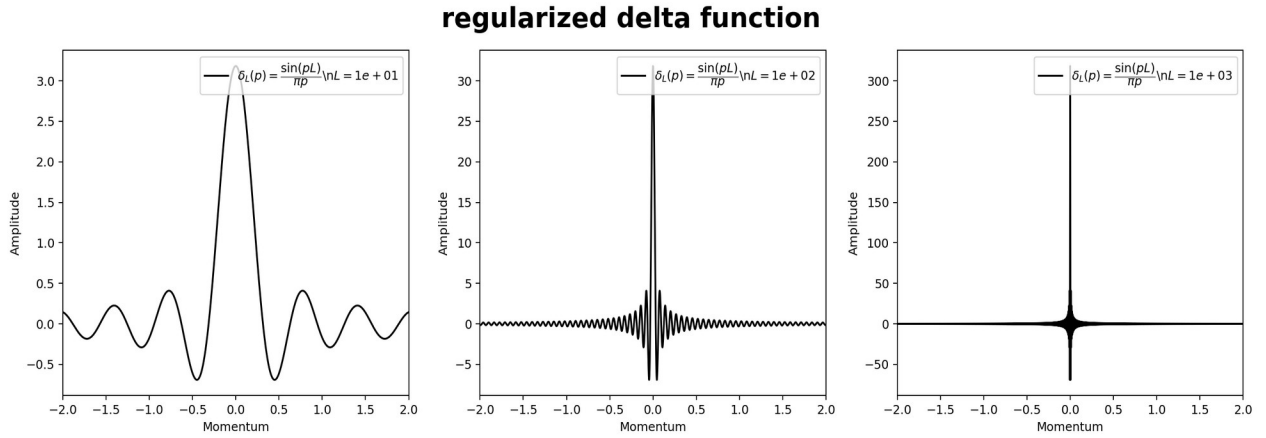


Figure 4. The regularized delta function for $L=10, 100$, and 1000 . Increasing L concentrates the distribution more strongly around $p=0$.

3. Part 3 - Stationary scattering and resonances

For the interacting calculations, the potential is

$$V(x) = (0.5x^2 - 0.8) \exp(-0.1x^2).$$

This is a symmetric central well surrounded by two barriers.

3.1 Numerical method

For each $E > 0$, the physical stationary state for a particle incident from the left has the asymptotic form below, with $k = \sqrt{2E}$ in the chosen units.

$$\begin{aligned} \psi_E(x \rightarrow -\infty) &= \exp(ikx) + R(E) \exp(-ikx) \\ \psi_E(x \rightarrow +\infty) &= T(E) \exp(ikx) \end{aligned}$$

For the finite-difference calculation, a modified state with unit transmitted amplitude is used instead:

$$\begin{aligned} \bar{\psi}_E(x \rightarrow -\infty) &= A(E) \exp(ikx) + B(E) \exp(-ikx) \\ \bar{\psi}_E(x \rightarrow +\infty) &= \exp(ikx) \end{aligned}$$

On an equally spaced spatial grid, the finite-difference Schrödinger equation can be written as

$$\bar{\psi}_E(x_{n-1}) = \left[2 + 2(\Delta x)^2 (V(x_n) - E) \right] \bar{\psi}_E(x_n) - \bar{\psi}_E(x_{n+1}).$$

Starting from two grid points in the right free region, this relation propagates the modified state step by step toward the left. In the left free region, $A(E)$ and $B(E)$ are obtained from the numerical wavefunction.

$$\psi_E(x) = \frac{\bar{\psi}_E(x)}{A(E)}, T(E) = \frac{1}{A(E)}, R(E) = \frac{B(E)}{A(E)}.$$

The physical transmission and reflection probabilities are $|T|^2$ and $|R|^2$, with $|T|^2 + |R|^2 = 1$ as the numerical consistency check.

3.2 Transmission spectrum

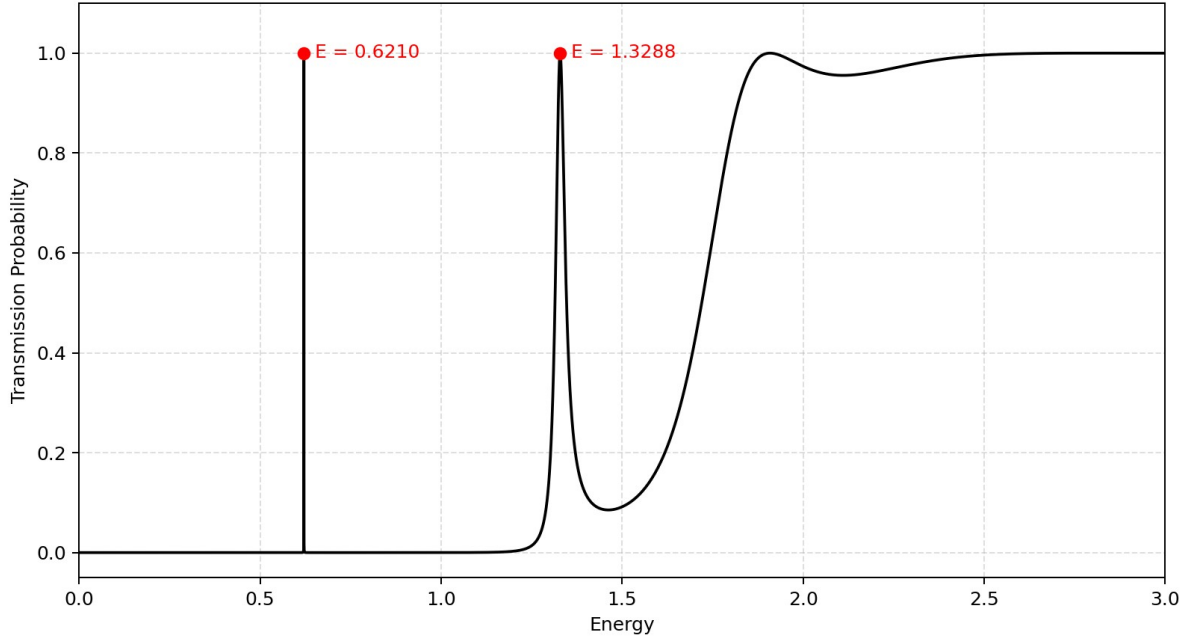


Figure 5. Calculated transmission probability over $0 < E < 3$. Two sharp below-barrier resonances are visible at $E \approx 0.6210$ and $E \approx 1.3288$; at higher energy the transmission tends toward unity.

The transmission spectrum shows two narrow resonance peaks at $E \approx 0.6210$ and $E \approx 1.3288$, where the transmission probability reaches almost one. At higher energies, the transmission gradually approaches one.

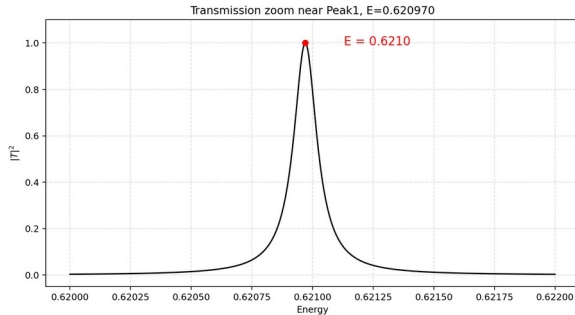


Figure 6a. Zoom of the first peak at $E \approx 0.6210$

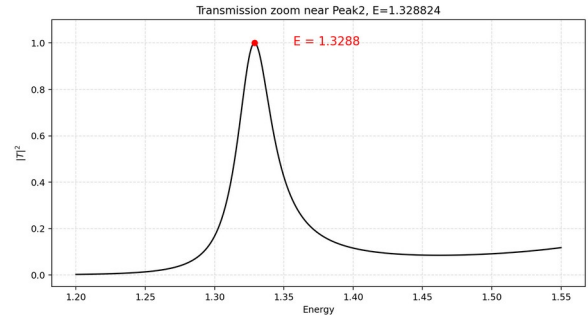


Figure 6b. Zoom of the second peak $E \approx 1.3288$

3.3 Continuum wavefunctions around the resonances

The following panels show the modified state $\bar{\psi}$ defined in Section 3.1.

First resonance. At $E=0.40$ and $E=0.75$, the physical transmission is almost zero and the wave is almost completely reflected. At the resonance energy $E \approx 0.62097$, $|T|^2 \approx 1$ and the wave is almost perfectly transmitted through the potential, and a large oscillation is also formed inside the central well.

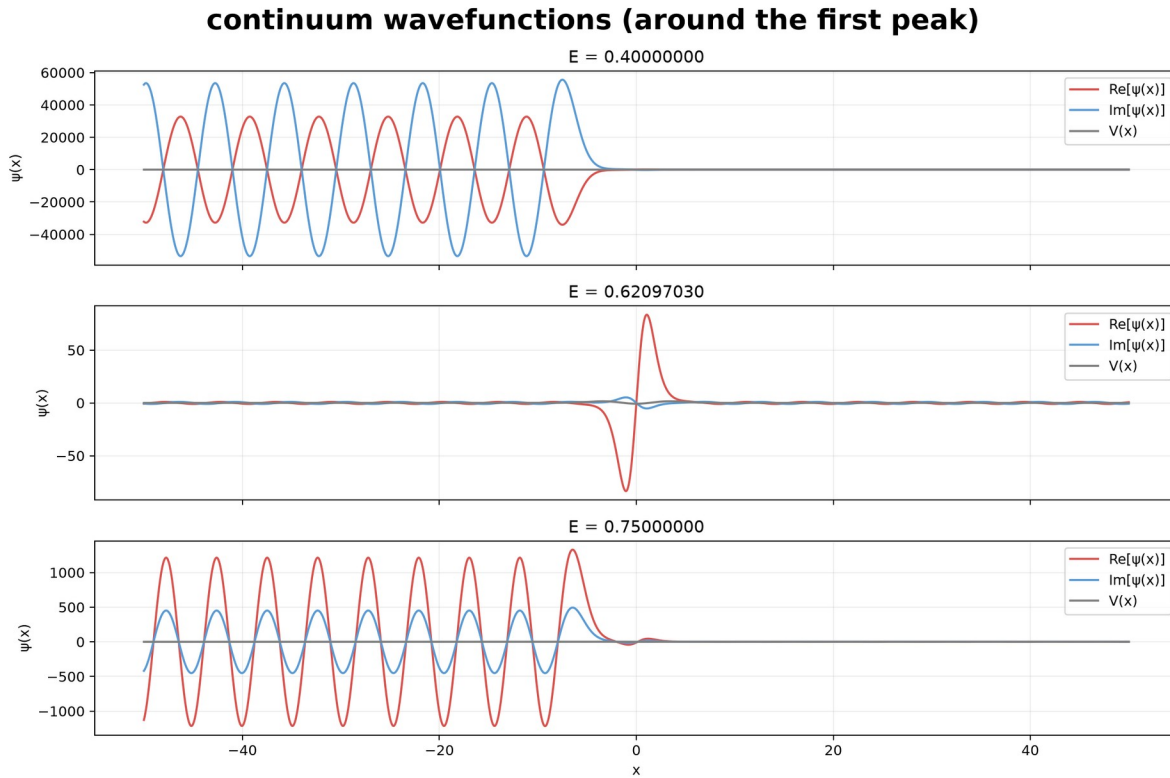


Figure 7. Modified continuum wavefunctions around the first resonance.

Second resonance. The same behavior appears around $E \approx 1.32882$. Away from the peak ($E=1.00$ and $E=1.60$), the transmission is much smaller, whereas at the resonance $|T|^2 \approx 1$ and the wave is almost perfectly transmitted through the potential, and a large oscillation is also formed inside the central well.

continuum wavefunctions (around the second peak)

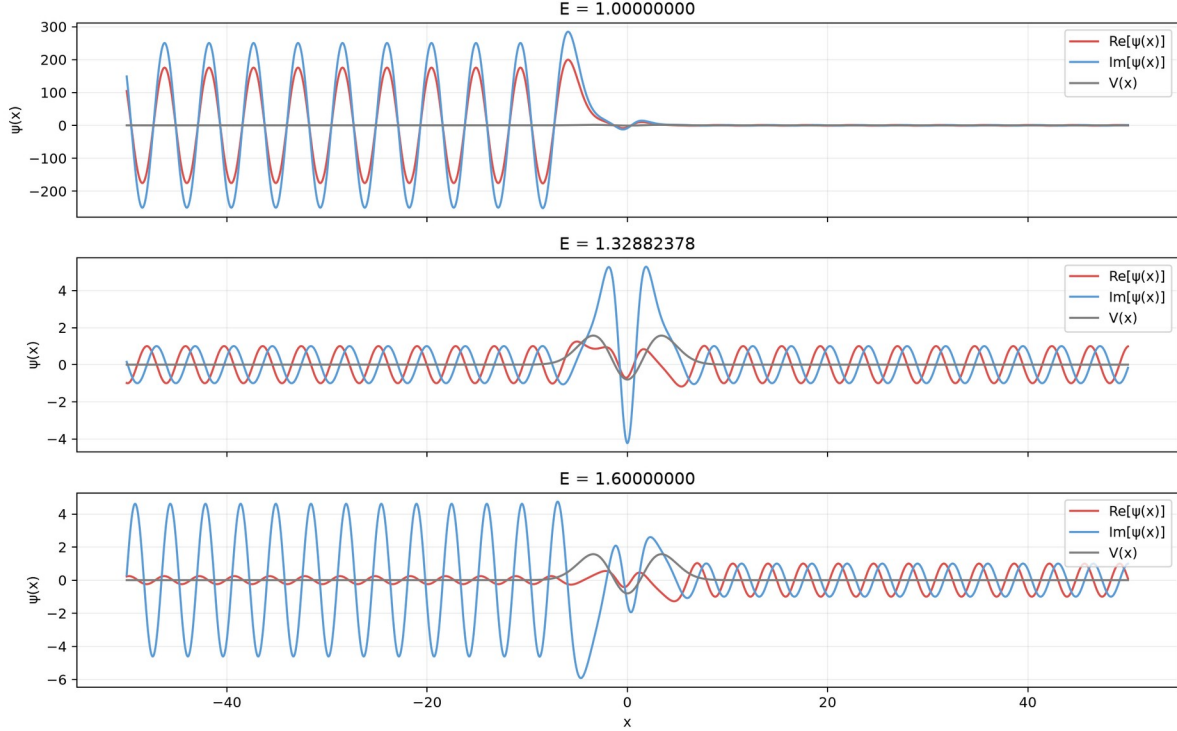


Figure 8. Modified continuum wavefunctions around the second resonance.

4. Part 4 - Wavepacket scattering

In Part 4, a Gaussian wavepacket is used as the incident wavepacket. It is constructed by superposing the left-incident stationary scattering states obtained in Part 3 with a Gaussian momentum distribution:

$$\Psi(t, x) = \int \phi(p) \psi_p(x) \exp(-i E(p)t) dp, E(p) = \frac{p^2}{2}.$$

Here $\phi(p)$ is the momentum-space coefficient of the incident Gaussian,

$$\phi(p) = \frac{1}{\sqrt{2\pi}} \int \Psi(0, x) \exp(-i p x) dx.$$

The function $\psi_p(x)$ is the left-incident stationary scattering state calculated in Part 3, while $\exp[-iE(p)t]$ gives the time evolution of each momentum component. The Gaussian distribution $|\phi(p)|^2$ is centered at p_0 , and its width determines which part of the transmission spectrum is included in the wavepacket. For the Gaussian convention used in Part 1, $\Delta p = \sqrt{\alpha}$ under $\hbar = 1$.

4.1 Blocked packet

The blocked packet is centered at $E_0 = 1.0$, well below the barrier top and away from either resonant peak. Across the relevant momentum distribution the transmission is extremely small. Therefore, almost the entire wavepacket is reflected back to the left. The snapshots show exactly this: the packet approaches the barrier, overlaps the potential well region, and then comes back as a reflected packet with negligible transmitted weight.

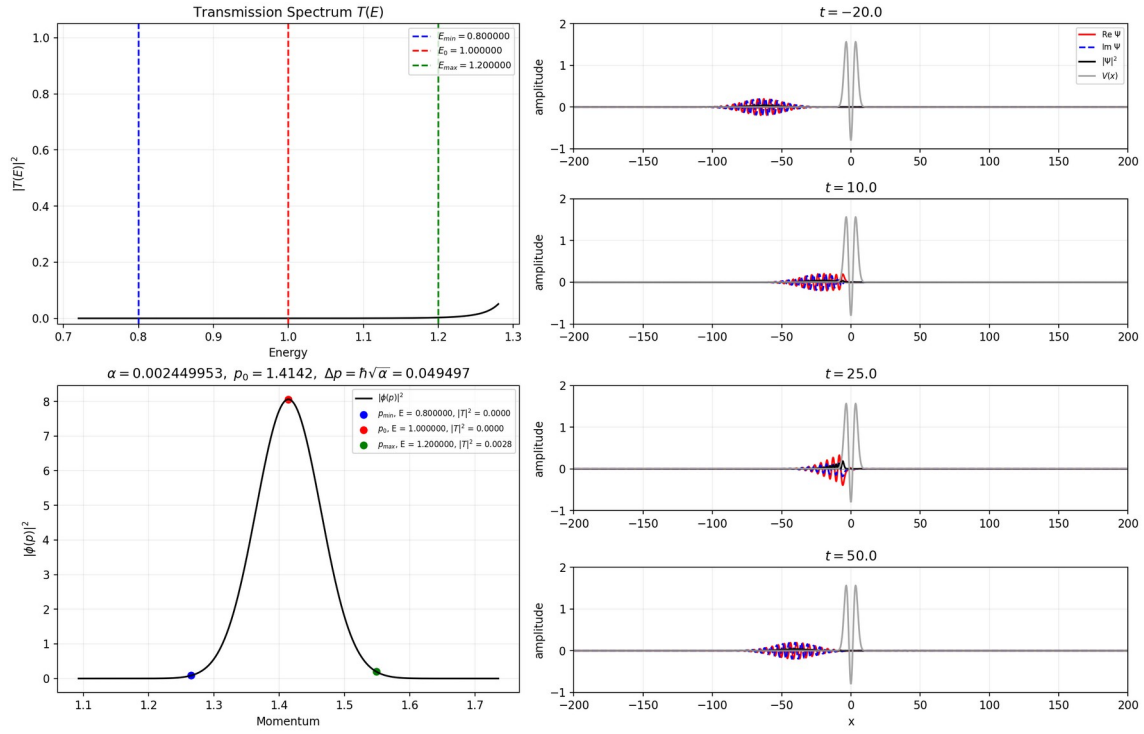


Figure 9. Blocked scenario. Left: stationary transmission spectrum and incident momentum density. Right: time snapshots showing almost complete reflection.

4.2 Partially blocked packet

The reference center $E_0 = 1.328814$ lies essentially on the second resonance peak, but the packet has a much larger finite momentum width $\Delta p = 0.015556$. Consequently only part of the incident spectrum sits in the high-transmission resonance region; neighboring components have much smaller $|T|^2$. Therefore, only the momentum components whose energies lie within the second resonance peak can be transmitted efficiently. Components whose energies deviate too far from the resonance have much smaller transmission probabilities and are reflected back to the left.

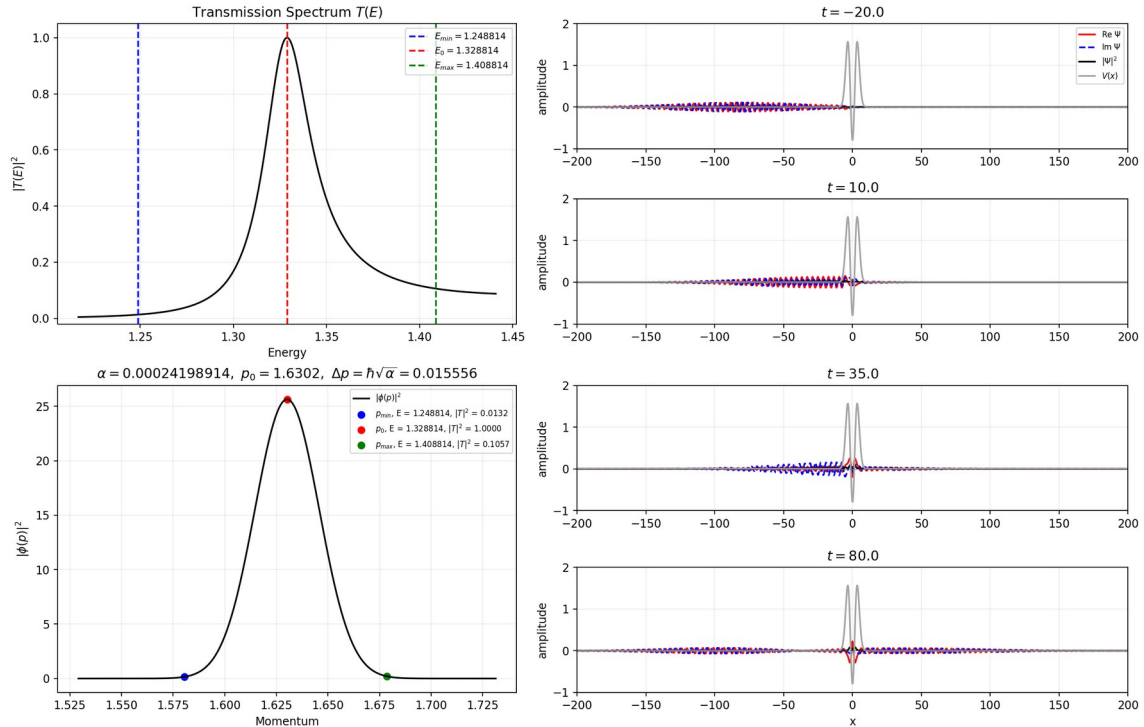


Figure 10. Partially blocked scenario centered near the second resonance. The finite spectral width covers both resonant and non-resonant energies, producing simultaneous transmission and reflection.

4.3 Passing packet

For $E_0=2.7$, the transmission probability is close to one over almost the whole momentum distribution. Therefore, almost all components of the wavepacket pass through the potential region and continue propagating to the right, with only a very small reflected part.

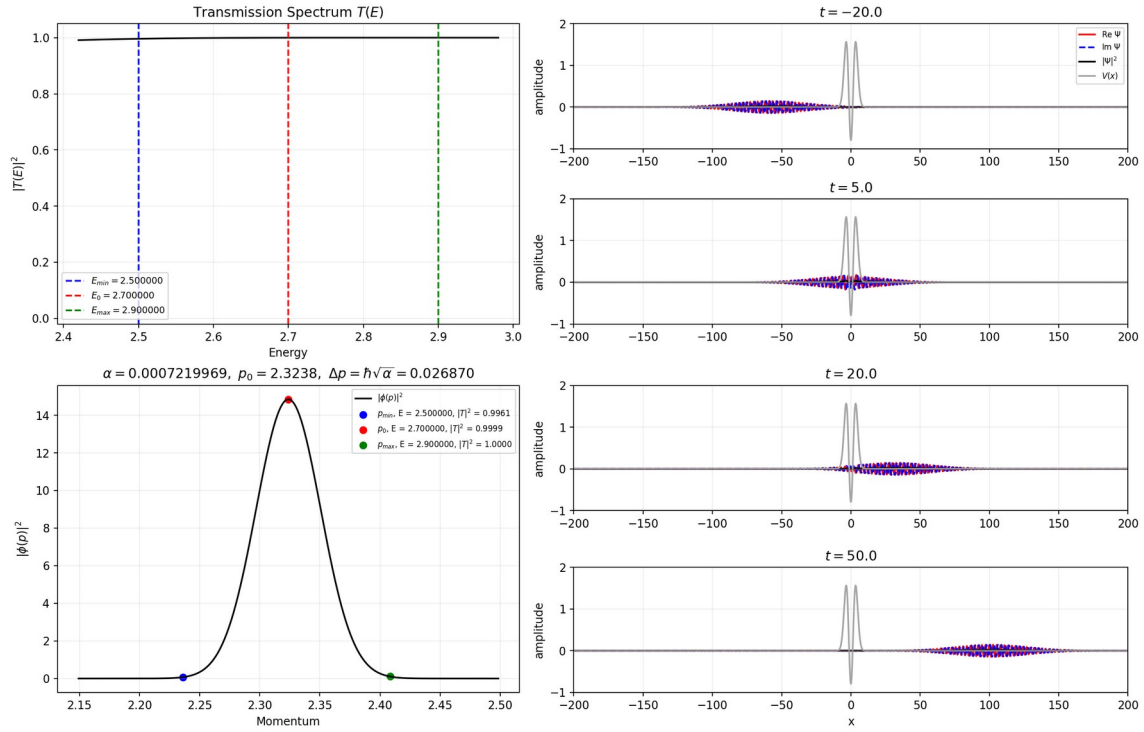


Figure 11. Passing scenario at high energy. Nearly the entire wavepacket crosses the potential.

4.4 Narrow range at first resonance peak

This case uses an extremely narrow momentum range centered on the first resonance peak. Since almost the entire range lies within the high-transmission region, the transmission probability is close to one for nearly all components of the wavepacket. Therefore, most of the wavepacket passes through the potential.

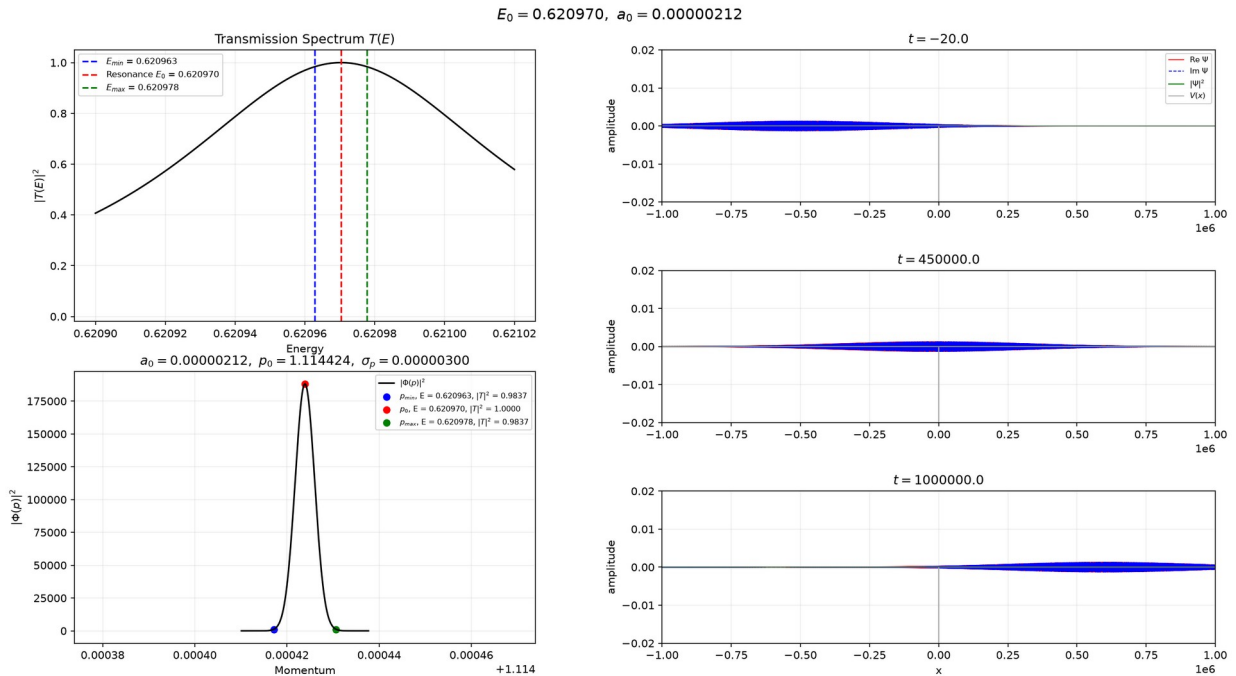


Figure 12. First-resonance packet centered at $E_0=0.62097030$. The three full-range snapshots show the real and imaginary parts before, during and after transmission.

4.5 Narrow range at second resonance peak

This case uses a very narrow momentum range centered on the second resonance peak. Since almost the entire range lies within the high-transmission region, the transmission probability is close to one for nearly all components of the wavepacket. Therefore, most of the wavepacket passes through the potential.

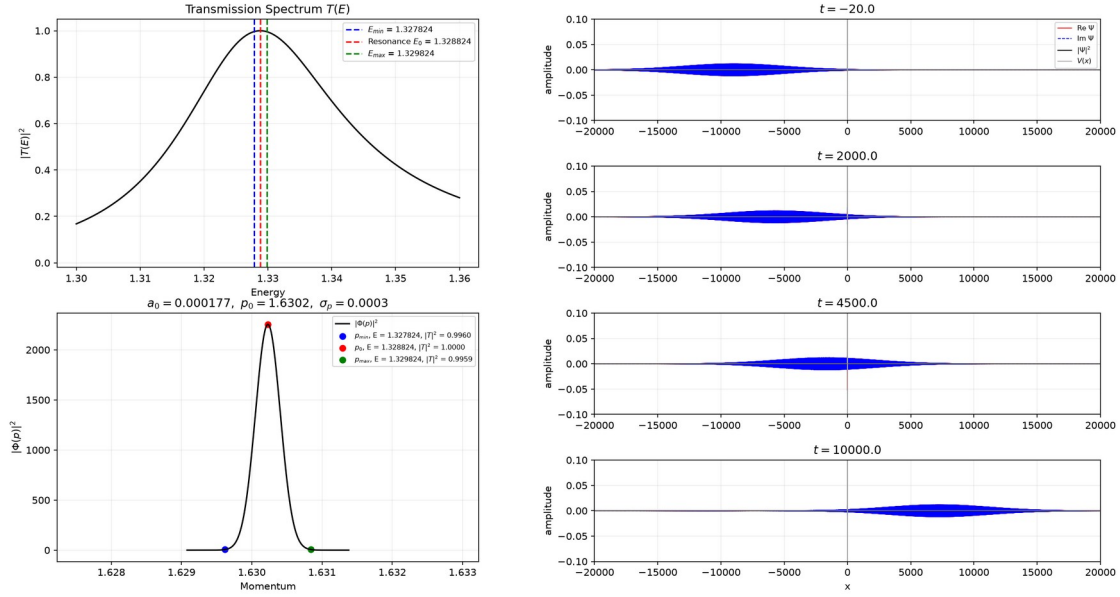


Figure 13. Extremely narrow-band packet centered exactly at the numerically determined second resonance. Its large spatial width is the Fourier counterpart of the tiny momentum width.

5. Part 5 - Box-basis eigenstates

In Part 5, the stationary Schrödinger equation is written as a matrix eigenvalue problem in a finite box. The particle-in-a-box basis is

$$\varphi_j(x) = \sqrt{\frac{2}{L}} \sin \left[\frac{j\pi \left(x + \frac{L}{2} \right)}{L} \right], -\frac{L}{2} \leq x \leq \frac{L}{2}.$$

The eigenfunctions are expanded in this basis as

$$\psi_n(x) = \sum_{j=1}^J c_j^{(n)} \varphi_j(x).$$

For $m=\hbar=1$, the Hamiltonian matrix contains a kinetic-energy part and a potential-energy part,

$$H_{ij} = T_{ij} + V_{ij}.$$

For the sine basis, the kinetic-energy part is diagonal:

$$T_{ij} = \frac{j^2 \pi^2}{2L^2} \delta_{ij}.$$

This is because the sine functions are orthogonal eigenfunctions of the particle-in-a-box kinetic-energy operator. The potential-energy part is

$$V_{ij} = \int_{-\frac{L}{2}}^{\frac{L}{2}} \varphi_i(x) V(x) \varphi_j(x) dx,$$

which can connect different basis functions.

The complete Hamiltonian matrix is then diagonalized,

$$H c^{(n)} = E_n c^{(n)},$$

to obtain the energies and the coefficients of each eigenfunction.

The finite box changes the positive-energy continuum into a discrete set of eigenvalues.

Here J is the number of basis functions, L is the box length, and N is the number of x points used for the numerical integrals.

5.1 Harmonic-oscillator validation

First, the method is tested with the harmonic oscillator $V(x)=x^2/2-0.8$, whose exact energies are $E_n=n-0.3$. The calculated energies are close to these values, showing that the box-basis method gives the expected spectrum.

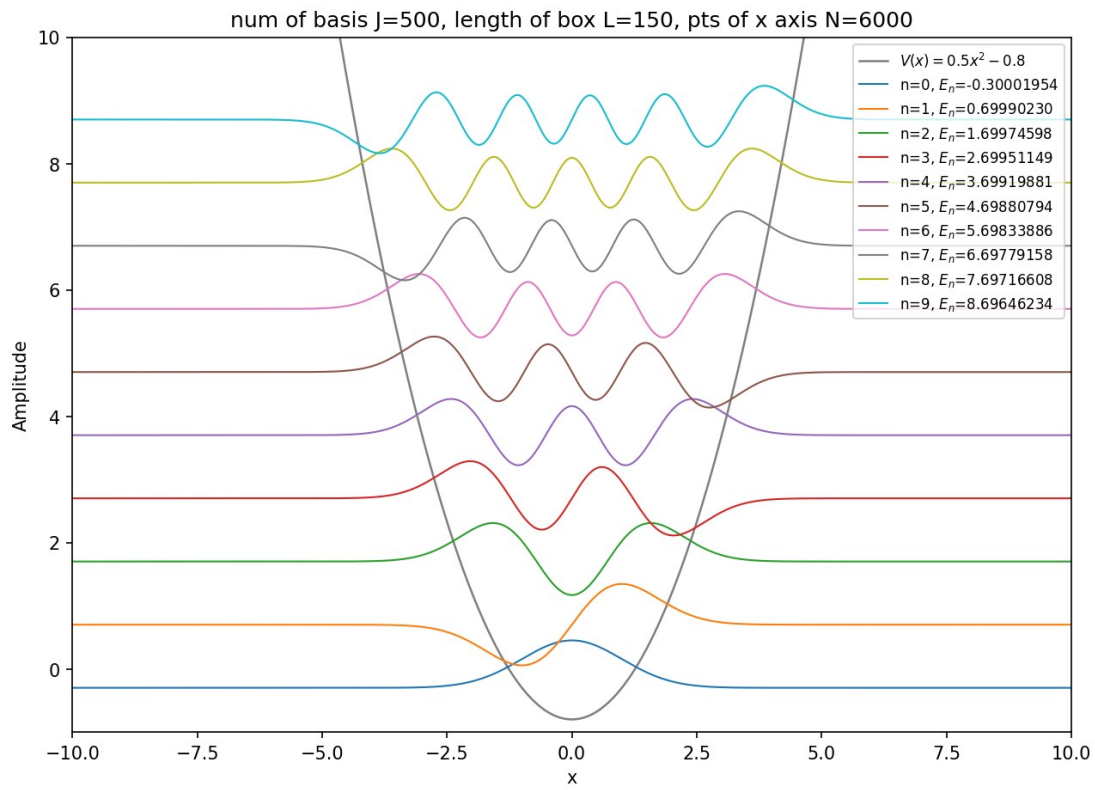


Figure 14. Harmonic-oscillator validation with $J=500$, $L=150$ and $N=6000$. The first ten energy-shifted eigenfunctions are shown with the potential.

5.2 First eigenstates of the short-range potential

For the short-range potential, $J=600$, $L=200$ and $N=10000$ are used. The lowest eigenvalue is $E_0=-0.29795964$. Its wavefunction is concentrated around the central well, so it is a bound state. The next ten eigenvalues are positive, and their wavefunctions extend across the box. Therefore, they are discretized continuum states rather than additional bound states.

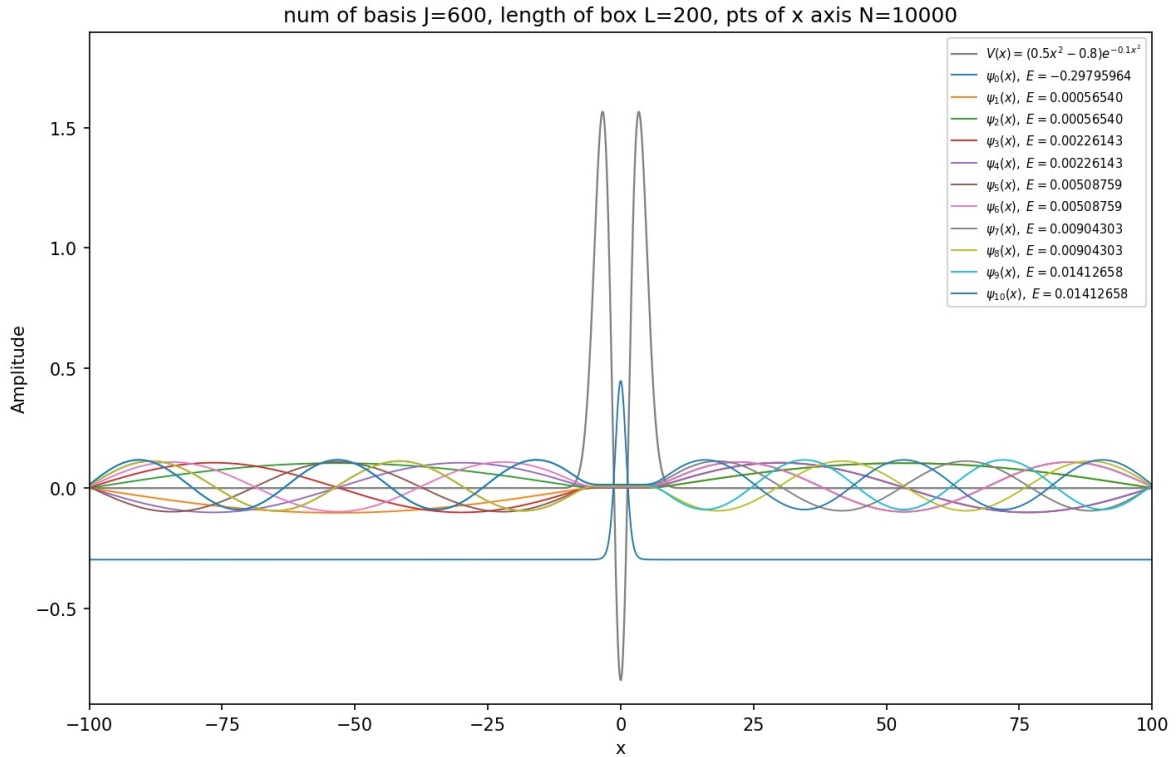


Figure 15. First 11 box eigenstates of the short-range potential. The negative-energy ground state is localized, while the positive-energy states extend through the finite box.

The same states are shown separately below. The number of oscillations increases with energy, whereas the negative-energy ground state remains localized near $x=0$.

num of basis $J=600$, length of box $L=200$, pts of x axis $N=10000$

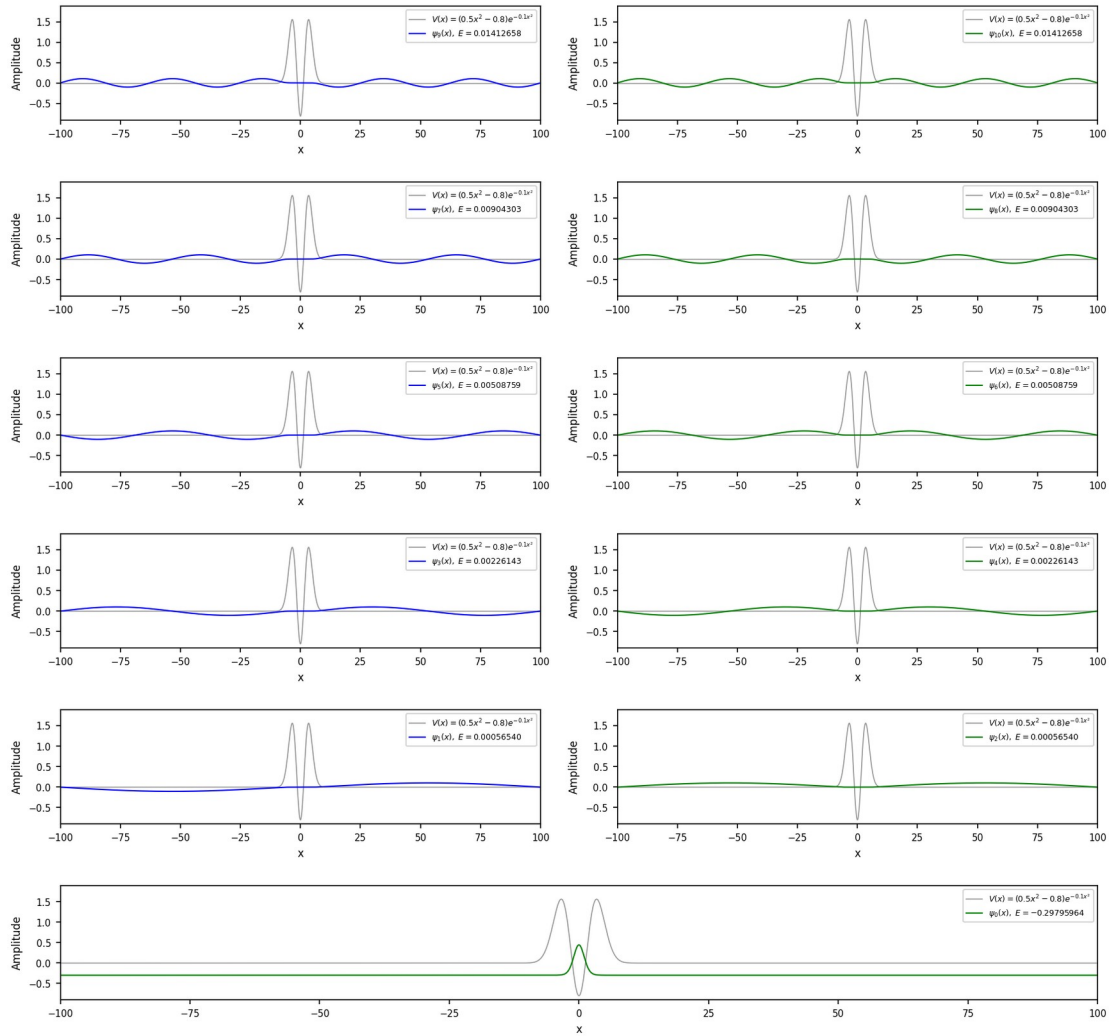


Figure 16. The first 11 eigenstates shown separately. The positive-energy states oscillate across the box and vanish at the two box boundaries.

5.3 All box eigenstates

With $J=201$, the calculation produces 201 discrete box eigenstates. Only the lowest state has $E < 0$. The many states with $E > 0$ fill the finite box and represent the scattering continuum after the box boundary has made it discrete.

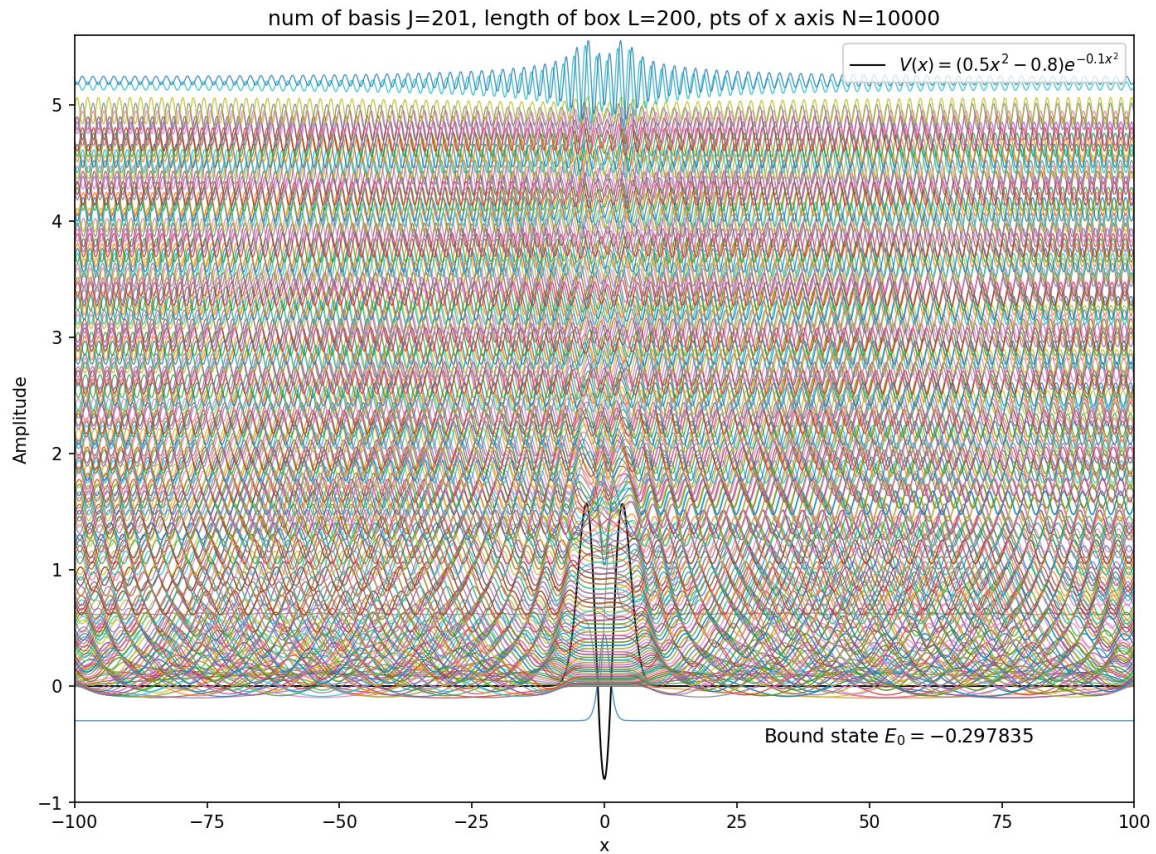


Figure 17. All 201 finite-box eigenstates for $L=200$. One negative-energy bound state lies below the dense set of positive-energy box states.

5.4 Bound state and resonance-like box states

Three states are selected from the larger $J=2000$ calculation. The first is the bound state at $E=-0.29795964$. The other two have $E=0.62097222$ and $E=1.33028105$, close to the two transmission peaks from Part 3 at $E=0.62097030$ and $E=1.32882395$. Thus, two positive-energy box states are found close to the resonance energies obtained in Part 3.

The state near the first resonance is more concentrated around the central region, while the state near the second resonance has larger oscillatory tails outside the potential.

However, the two $E>0$ states are not bound states. Their wavefunctions still have oscillatory tails outside the potential, and their exact box eigenvalues change when L is changed.

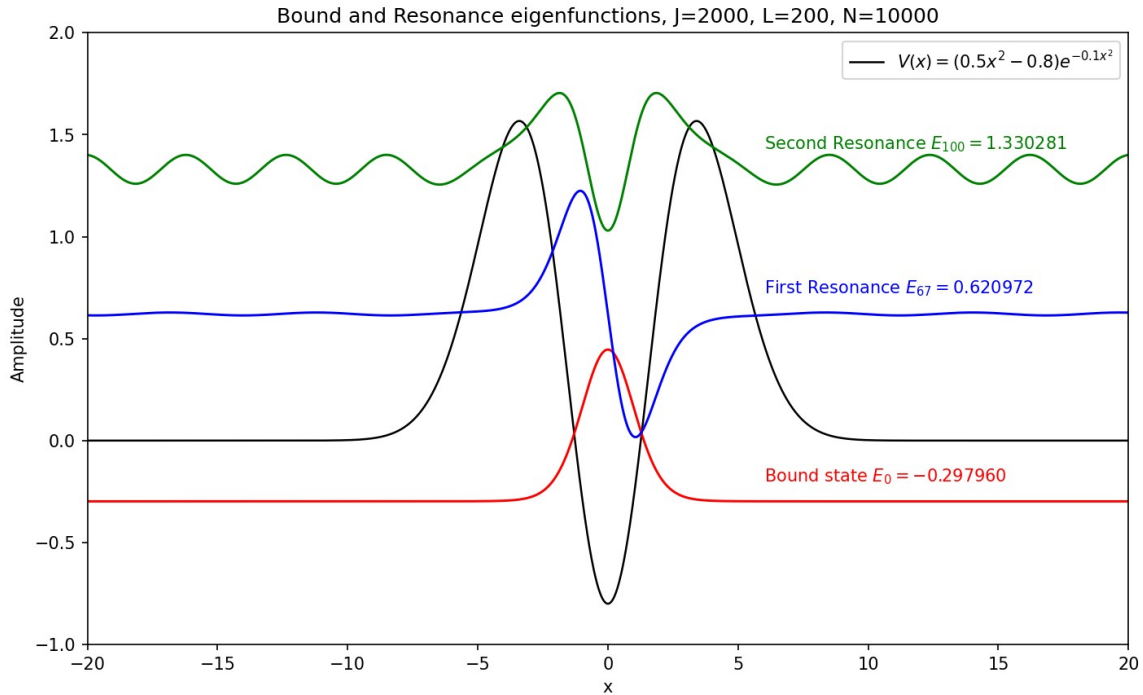


Figure 18. The bound state and two positive-energy box eigenstates selected near the Part 3 resonance energies. The first resonance-like state is more strongly concentrated around the central region.

6. Part 6 - Complex scaling and resonance states

Part 6 uses the box-basis method from Part 5 after rotating the spatial coordinate into the complex plane, $x \rightarrow x \exp(i\theta)$. For the same short-range potential, the complex-scaled Hamiltonian is

$$H_\theta = \exp(-2i\theta) T + V[x \exp(i\theta)] = -\frac{1}{2} \exp(-2i\theta) \frac{d^2}{dx^2} + V[x \exp(i\theta)].$$

The kinetic-energy matrix is multiplied by $\exp(-2i\theta)$, while the potential matrix is evaluated at the complex coordinate. The resulting non-Hermitian matrix is diagonalized in the same sine basis:

$$H_\theta c^{(n)} = E_n(\theta) c^{(n)}.$$

The calculation uses $J=2000$, $L=200$ and $N=10000$ for $\theta=0, 0.05, 0.10, 0.15, 0.20$ and 0.25 . As θ increases, the discretized continuum rotates downward in the complex-energy plane by approximately 2θ , whereas resonance eigenvalues remain nearly unchanged. These stable points can therefore be selected as complex resonance energies.

Each resonance energy is written as

$$E_n = E_{res,n} - i \frac{\Gamma_n}{2}, \Gamma_n = -2 \operatorname{Im}(E_n).$$

The real part $E_{res,n}$ gives the center of the transmission peak. The positive quantity Γ_n gives the full width at half maximum of an isolated Breit-Wigner peak:

$$|T(E)|^2 \approx \frac{\left(\frac{\Gamma_n}{2}\right)^2}{\left[(E - E_{res,n})^2 + \left(\frac{\Gamma_n}{2}\right)^2\right]}.$$

Thus, a pole closer to the real axis produces a narrower peak, while a pole with a larger negative imaginary part produces a broader peak. The same rotation also gives $\psi_\theta(x) = \psi[x \exp(i\theta)]$. For a sufficiently large θ , the outward growth of a Siegert resonance state becomes decay, so the state can be treated numerically in a finite basis.

6.1 Complex eigenvalue spectra

At $\theta=0$, the discretized continuum lies on the real-energy axis. As θ increases, these states rotate into the lower half of the complex-energy plane. A few isolated eigenvalues remain almost at the same positions; these are the resonance states.

complex eigenvalues (plotted in one picture for various values of theta)

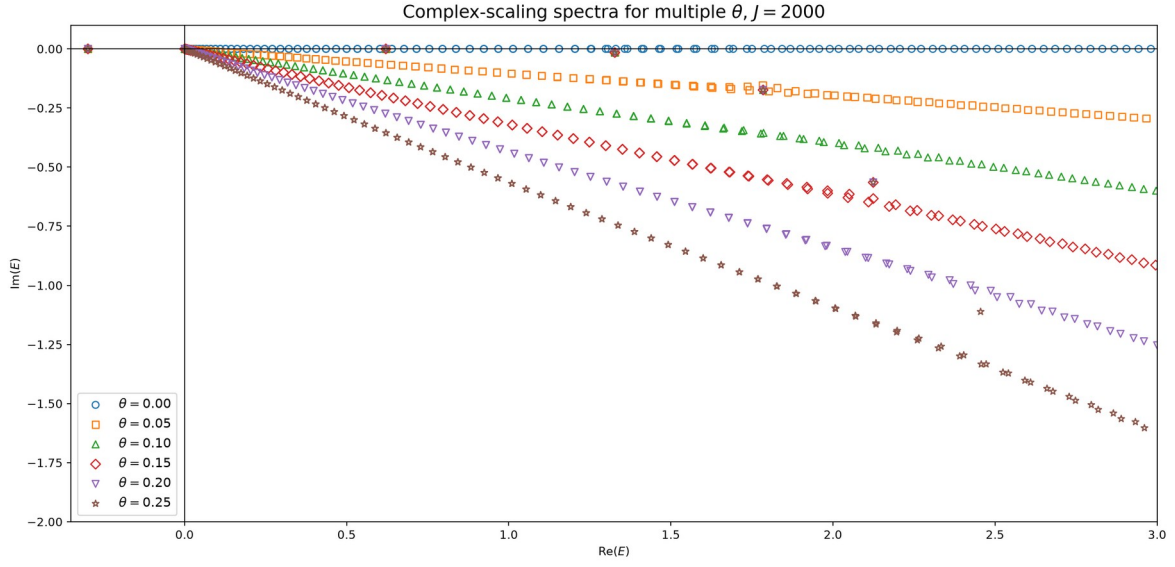


Figure 19. Complex-scaled eigenvalues for several values of θ . The continuum rotates downward, while isolated resonance eigenvalues remain nearly fixed.

complex eigenvalues (separately for each theta)

Complex-scaling individual spectra, $J = 2000$

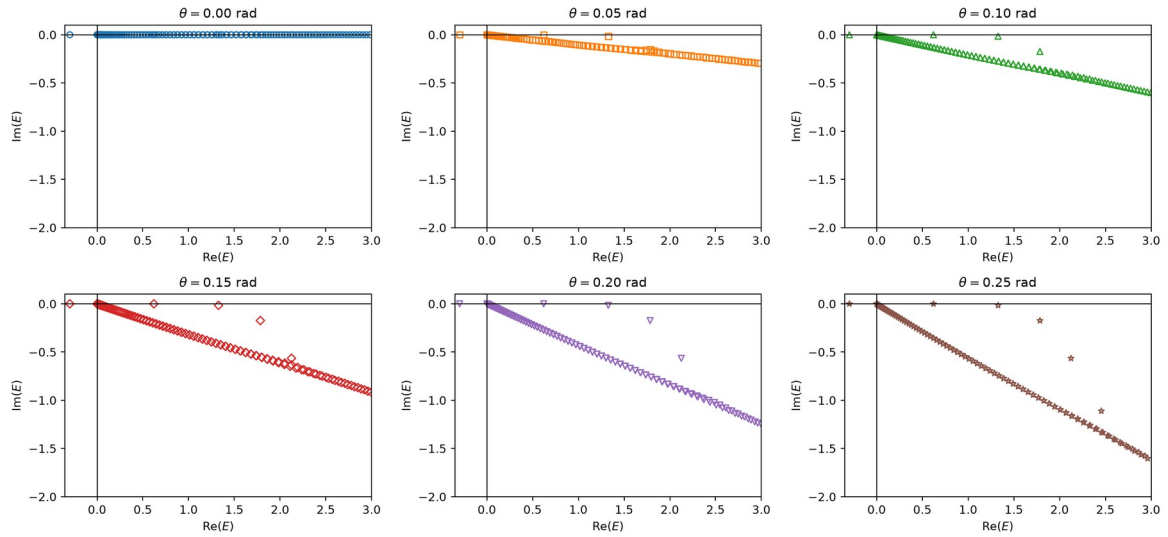


Figure 20. Individual panels for $\theta=0, 0.05, 0.10, 0.15, 0.20$ and 0.25 , making the motion of the continuum easier to follow.

The two lowest stable points are $E_1 \approx 0.620971 - 0.000058i$ and $E_2 \approx 1.327197 - 0.015447i$. Their real parts lie near the Part 3 transmission maxima. Using $\Gamma = -2 \text{Im}(E)$ gives $\Gamma_1 \approx 0.000116$ and $\Gamma_2 \approx 0.030894$. The much smaller Γ_1 explains why the first transmission peak is so sharp.

6.2 Breit-Wigner profiles

The complex resonance energies obtained above are substituted into the Breit-Wigner expression and compared with the Hermitian transmission probability calculated in Part 3.

Breit-Wigner profiles (all)

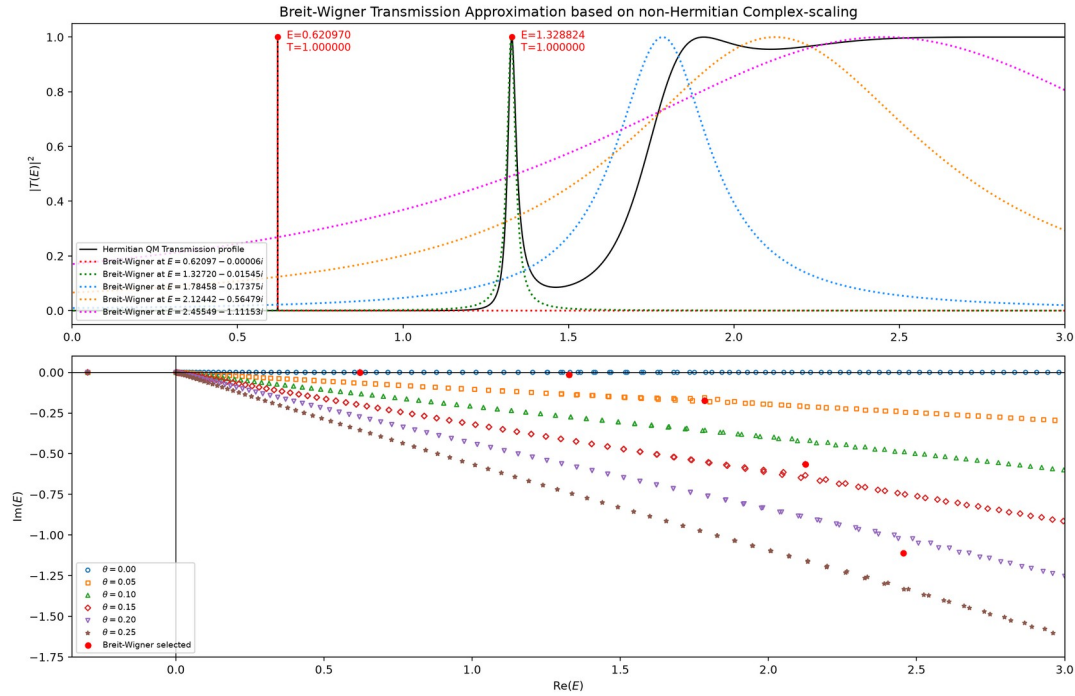


Figure 21. The Hermitian transmission profile compared with Breit-Wigner profiles obtained from the selected complex resonance energies.

Breit-Wigner profiles (detailed zoom)

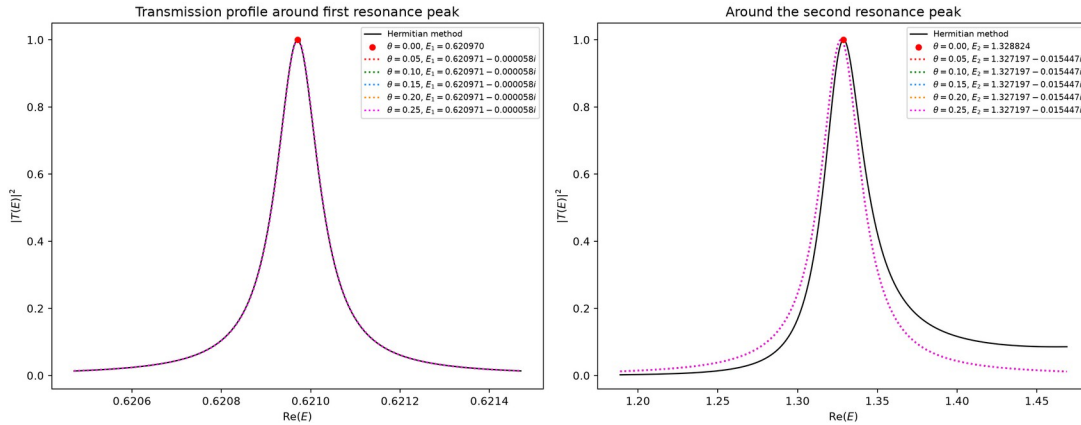


Figure 22. Detailed comparison around the first and second transmission peaks.

At the first resonance, the Breit-Wigner approximation nearly coincides with the Hermitian curve and reproduces its narrow line shape. For the second resonance, the center and overall width are still close, but the single-pole curve no longer follows the full profile away from the top. The remaining difference is attributed to the non-resonant background and the influence of broader resonances.

6.3 Resonance wavefunction

The last two figures show the second resonance state at $E_{res} \approx 1.327197 - 0.015447i$. The original Siegert state satisfies outgoing boundary conditions and grows at large $|x|$, so it is not square integrable. After the rotation $\psi_\theta(x) = \psi[x \exp(i\theta)]$ with $\theta = 0.1$, the same state decays away from the potential.

wavefunction - plot 1

Resonance state at $E_{res} = 1.327197 - 0.015447i$

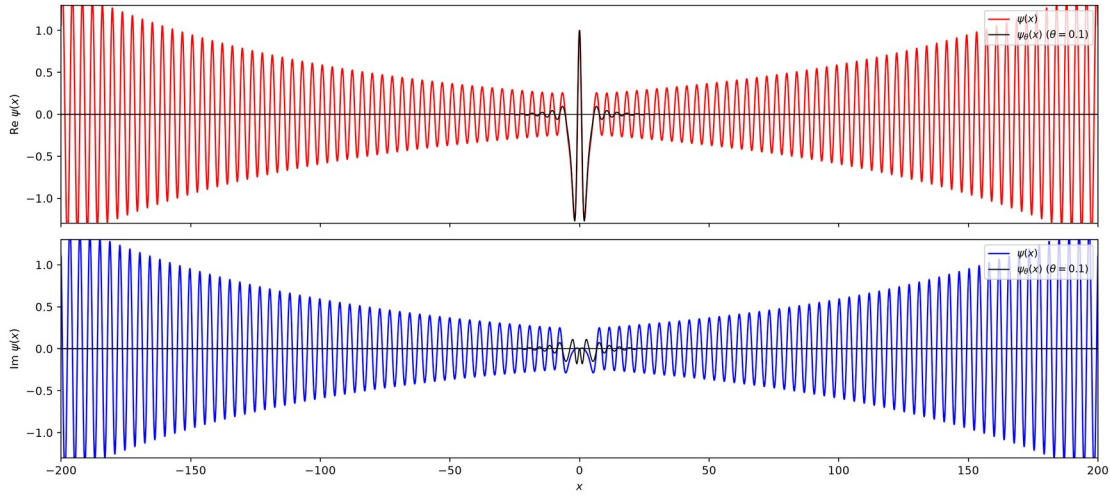


Figure 23. The second resonance wavefunction over $-200 \leq x \leq 200$ before and after complex scaling with $\theta = 0.1$.

wavefunction - plot 2

Resonance state at $E_{res} = 1.327197 - 0.015447i$

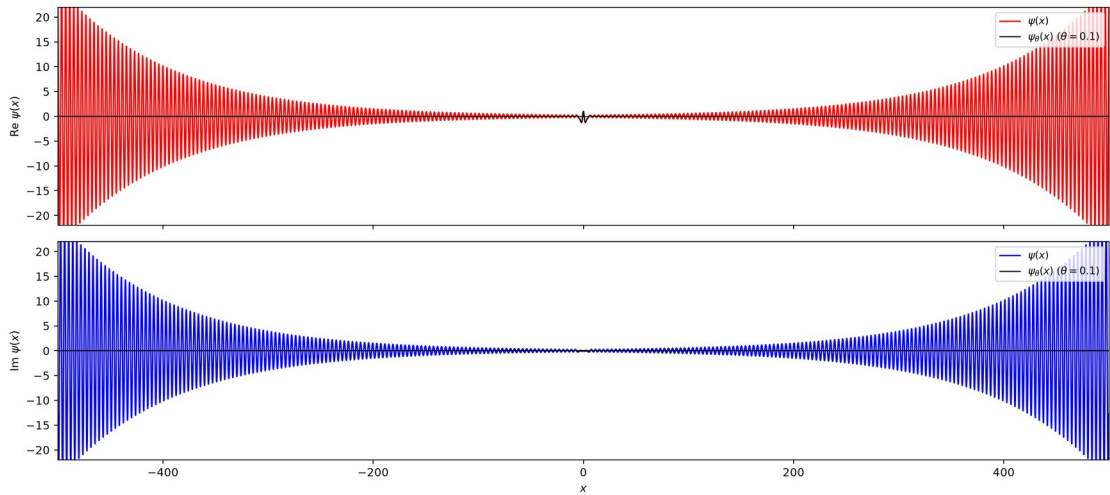


Figure 24. The same resonance state over $-500 \leq x \leq 500$, showing the outward growth of the unscaled state and the decay of the complex-scaled state.

The larger interval makes the different asymptotic behaviors clearer: the unscaled state continues to grow outward, whereas the complex-scaled state remains localized around the potential.

6.4 Comparison of resonance energies

The Hermitian scan in Part 3 gives peak energies $E_1 \approx 0.62097030$ and $E_2 \approx 1.32882395$, while complex scaling gives poles $0.620971 - 0.000058i$ and $1.327197 - 0.015447i$. Comparing the Hermitian maxima with the real parts of these poles, the first values almost coincide and the second remain close, so both methods locate the same two resonance regions. Complex scaling also gives $\Gamma_1 \approx 0.000116$ and $\Gamma_2 \approx 0.030894$; the small shift of the second center is consistent with its broader, less isolated line shape.